(O1), the relations case at $n = 2$: what is proved, and the right form of the problem

Notation as in `mechanism_note.tex`, Section 9. Throughout: $n = 2$ ($\nu = 3$), $2 \le b \le 3$ ($1 \le s \le \frac{3}{2}$, $r = 1/s$), class $\mathcal{C}$, and the *relations case*: every feature's interference sits on $T = \{1, 2\}$, so $\Phi_l(x) = \overline{\beta_l} + W_l \rho^{2l(x_1+x_2)}$. Put $v_l = |W_l| = |\beta_{l,12}|$, $\theta_l = \arg W_l$, $w = 2(x_1 + x_2) \in \mathbb{Z}_p$,

$$\varphi(w) = \sum_{l \in F} v_l \cos(2\pi l w/p + \theta_l), \qquad D_\delta(w) = \varphi(w) - \varphi(w - \delta), \qquad s_l(\delta) = |\sin(\pi l \delta/p)|.$$

$a_l = 3^{1-s} Q_l^s$ is the interference-free (IF) price, $L' = \{l : |\beta_l| > a_l\}$ the cheap set, δ_0 a minimiser of $(Ka)_\delta = \sum_l a_l \kappa_l(\delta)$, and $\|a\|_r = (\sum a_l^r)^s$. The IF value is $\ge \Gamma_s \|a\|_r$, and $(Ka)_{\delta_0} \le \Gamma_s \|a\|_r$. *Payment.* Part I's Theorem A with $B = C = 0$ gives $\operatorname{Re} \beta_l \le a_l + c\, v_l$ with $c = c_* := 1/(2\sqrt{3}) = 0.2887$, and $v_l \le Q_l^s = 3^{s-1} a_l$.

1. The oscillation identity

Lemma 1. *For every $\delta \ne 0$: $h \le \sum_l \kappa_l(\delta) \operatorname{Re} \beta_l - \frac{1}{2} \operatorname{osc}_w D_\delta(w)$, where* $\operatorname{osc} = \max - \min$.

Proof. $g(x, \pm\delta) = M_{\pm\delta} + D_{\pm\delta}(w)$ with $M_\delta + M_{-\delta} = 2 \sum_l \kappa_l \operatorname{Re} \beta_l$. Use *separate* minimisers for $\pm\delta$: $h \le \frac{1}{2}(M_\delta + M_{-\delta}) + \frac{1}{2}(\min D_\delta + \min D_{-\delta})$. Finally $D_{-\delta}(w) = \varphi(w) - \varphi(w + \delta) = -D_\delta(w + \delta)$, so $\min D_{-\delta} = -\max D_\delta$. □

Remark 2. The note's derivation evaluates $\pm\delta$ (and all directions of μ) at a *common* point and so meets the one-sided quantity $-\min \psi_\mu$ of a second difference. Lemma 1 replaces it by half the oscillation of the first difference, a *two-sided* quantity. D_δ has amplitudes $b_l = v_l |c_l(\delta)| = 2 v_l s_l$, the payment is $c \sum v_l \kappa_l = c \sum_l b_l s_l$, and $\mathbb{E}_w D_\delta^2 = \sum_l v_l^2 \kappa_l(\delta)$ exactly (the frequencies $l \in F$ are distinct and $l \not\equiv -l'$; no relation enters).

2. Parseval closes the relations case for $p \le 23$

Theorem 3 (Parseval closure)**.** *Let c be any valid single-T payment constant ($\operatorname{Re} \beta_l \le a_l + c v_l$). If $c^2 \sum_{l \in L'} \kappa_l(\delta_0) \le 1$, then $h \le$ IF value; if the inequality is strict, equality forces $L' = \emptyset$ (then the note's Proposition prop:nocheap gives the count). With $c = c_*$ the threshold is $\sum_{L'} \kappa_l(\delta_0) \le 12$.*

Proof. Mean zero and Bhatia–Davis give $\operatorname{osc} D \ge 2\sqrt{\mathbb{E} D^2}$. So by Lemma 1 and Cauchy–Schwarz, $h \le (Ka)_{\delta_0} + c \sum_{L'} v_l \kappa_l - \sqrt{\sum_{L'} v_l^2 \kappa_l} \le (Ka)_{\delta_0} + \sqrt{\sum_{L'} v_l^2 \kappa_l}\, \big(c \sqrt{\sum_{L'} \kappa_l} - 1\big)$. □

This is the note's Theorem thm:loc with the factor $2^n - 1$ replaced by the number of sets T that carry cheap interference (one, here); I record it because its consequences were not drawn:

Corollary 4. *(a) Since $\sum_{l \in F} \kappa_l(\delta) = p/2$ for every $\delta \ne 0$, **the relations case is closed for every** $p \le 23$, for every cheap set, every $b \in [2, 3]$, any width; maximisers in it are interference-free. (b) At every prime, the relations case is closed whenever $|L'| \le 6$ ($\kappa < 2$). (c) Cheap sets inside a programme support: for the supports computed ($p \le 31$ exact, $37 \le p \le 101$ heuristic, file `supports_s1.5.json`)* $\max_\delta \sum_{l \in S} \kappa_l(\delta) \le 10.9 < 12$, *so every such cheap set is closed in* every *direction (in particular at whichever binding direction the network makes δ_0). (d) General T at $n = 2$: the note's threshold 4 gives **all of class $\mathcal{C}$ at $n = 2$ for** $p \in \{5, 7\}$ ($p/2 \le 3.5$). Alternatively (better when features use few sets T), with $A_T = \sum_{l \in L', \beta_{l,T} \ne 0} \kappa_l(\delta_0)$, the bound holds when $\sum_T \sqrt{A_T} \le 2\sqrt{3}$ (use $R_l \le \sum_T |\beta_{l,T}|$ and Bhatia–Davis per T).*

3. The sharp single-T payment, and $p \le 59$

The extremal family of the relations note (two aligned atoms) has bonus $(1-(2/3)^{s-1})|D|$. Let $c_{12}(s) := 1-(2/3)^{s-1}$ (0.1835 at $b=2$).

Proposition 5 (P_{12} for sign atoms)**.** *Let $\beta_i \in \mathbb{C}$, $\xi_i, \eta_i \in \{\pm 1\}$, $B = \sum \beta_i \xi_i = 0$, $C = \sum \beta_i \eta_i = 0$, $A = \sum \beta_i$, $D = \sum \beta_i \xi_i \eta_i$. Then for $1 \le s \le \frac{3}{2}$: $|A| - c_{12}(s)|D| \le 3^{1-s}(\sum_i |\beta_i|^r)^s$, with equality for the aligned pair.*

Proof. Group the atoms by (ξ, η) with sums $G_{\pm\pm}$; merging a group lowers $\sum |\beta|^r$ and keeps A, B, C, D. $B \pm C = 0$ give $G_{++} = G_{--} =: g$, $G_{+-} = G_{-+} =: h$, so $A = 2(g+h)$, $D = 2(g-h)$, cost $\ge 2|g|^r + 2|h|^r$. For fixed moduli $|A| - c|D|$ is largest when g, h are aligned. With $|h| = t|g|$, $t \le 1$, $|g| = 1$ and $1 - c_{12} = (2/3)^{s-1} = 3^{1-s}2^{s-1}$:

$$\tfrac{1}{2}\big[(|A| - c|D|) - 3^{1-s}(2+2t^r)^s\big] = (2/3)^{s-1}\big[1-(1+t^r)^s\big] + t(1+c) \le t^r\big[(1+c) - s(2/3)^{s-1}\big]$$

by Bernoulli and $t \le 1$. The bracket is $2 - (1+s)(2/3)^{s-1} \le 0$, because $\log((1+s)(2/3)^{s-1})$ is concave in s, equal to $\log 2$ at $s = 1$ and to $\log 2.04$ at $s = \frac{3}{2}$. □

Numerically (SLSQP, $m \le 5$ complex atoms, 150 starts each, `c12.py`) the supremum of $(|A| - 3^{1-s}Q^s)/|D|$ under $B = C = 0$ with *complex* unit ξ, η is again $c_{12}(\frac{3}{2}) = 0.1835$.

Conjecture 6 (P_{12})**.** *Proposition 5 holds for complex unit ξ_i, η_i.*

Corollary 7. *With $c = c_{12}$ the Parseval threshold is $1/c_{12}^2 \ge 29.7$ on $1 \le s \le \frac{3}{2}$. Hence the relations case is closed for every $p \le 59$ (and for $|L'| \le 14$ at every prime) for networks whose input Fourier phases at each feature are signed ($\hat{u}_{it}(l) \in \mathbb{R} \cup i\mathbb{R}$), and for all networks if (P_{12}) holds.*

So for $p \le 59$ the relations case is a *single-feature* statement. (P_{12}) is an $n = 2$ atom inequality of the kind Theorem A proves; the spectral form of the cost lemma in the note is a natural route.

4. $p > 23$: what remains, and numerical status

For $p > 23$ Parseval can fail only if the cheap set carries more than 12 units of $\kappa(\delta_0)$, i.e. contains most of F. Maximising over $v \in [0, \tau a]$, $\tau = 3^{s-1}$, gives a criterion free of phases and of v:

$$(C_{L2}) \qquad \widetilde{H}(a) := \min_\delta \big[(Ka)_\delta + \tau g_\delta(a)\big] \le \Gamma_s \|a\|_r \quad \forall a \ge 0, \qquad g_\delta(a) = \max_{0 \le z \le a} \big(c_* \textstyle\sum z_l \kappa_l - \sqrt{\sum z_l^2 \kappa_l}\big).$$

Water-filling: $g_\delta = c_* \lambda\big(\sum \kappa_l m_l - 12\big)$, $m_l = \min(a_l/\lambda, 1)$, $\sum \kappa_l m_l^2 = 12$ ($g_\delta = 0$ if $\sum_{a_l > 0} \kappa_l \le 12$). $\widetilde{H}$ is concave and 1-homogeneous; (C_{L2}) implies the IF bound in the relations case. *Numerics* (CMA-ES, many starts, `cl2f.py`, `cl2g.py`; $b = 2$). On the region where the Parseval gain is substantial ($\tau g \ge 0.01\,\Phi$) the supremum of $\widetilde{H}/(\Gamma_s \|a\|_r)$ is $0.802, 0.827, 0.831, 0.840, 0.837, 0.833, 0.831, 0.844, 0.839$ for $p = 29, 31, 37, 41, 43, 47, 53, 59, 61$ (larger p in `cl2g.out`); the maximisers spread over 11–16 frequencies. Uniform allocations on F give 0.66–0.76 for $29 \le p \le 97$. So spread cheap sets are beaten by coverage loss with $\ge 15\%$ to spare. (These are lower estimates of suprema from local search, i.e. evidence; Γ_s for $p > 31$ is the heuristic value.) This is the precise sense in which "coverage" enters: not as a property of the cheap set, but as the concavity loss of an allocation spread enough to defeat Parseval.

A crude scaling for where (C_{L2}) must eventually fail: a uniform allocation on $k \ge 12$ frequencies with coverage γ has ratio $\approx (1.5\gamma/\sqrt{k} - 1.73\sqrt{\gamma}/k)/\Gamma_s$; with $\Gamma_s \approx \gamma^\star/\sqrt{K^*}$ this exceeds 1 only once $K^* \gtrsim 16$, i.e. $\ln p \gtrsim 11$. Below that, Parseval plus coverage suffices; beyond it an extreme-value (not L^2) bound is needed.

5. (O1) as stated in the note is false; the correct statement

- The plain one-sided constant $\sigma(X) = \min_{b,\theta} \max_w (-\sum b_l \cos(2\pi lw/p + \theta_l))/\sum b_l$ of the full set is $\sigma(F) = 1/(p-1)$, attained at *zero* phases: $\sum_{l\in F} \cos(2\pi lw/p) = \frac{1}{2}(p\,\mathbf{1}_{w=0} - 1)$ is a spike. So good coverage (here: perfect coverage) produces a *small* one-sided constant.
- For the programme supports themselves (`sigma_supports.out`): $\sigma = 0.250, 0.167, 0.602, 0.337, 0.478, 0.516, 0.$ for $p = 5, 7, 11, \ldots, 31$. Below $\frac{1}{2}$ except at $p = 11, 19$, and below c_* at $p = 5, 7$ (where $\operatorname{spt} y^\star = F$).
- Yet the relations case is closed at all these primes (Corollary 4). The plain one-sided inequality is neither necessary nor the right object: the network faces the oscillation of a *first difference* (Lemma 1), and a spike φ has $\operatorname{osc} D_\delta \approx 2 \max \varphi$, the full amplitude.

Corrected (O1′). For all $a \geq 0$, $v \in [0, \tau a]$ and phases, $\min_{\delta\neq 0} \left[\sum_l \kappa_l(\delta)(a_l + c\,v_l) - \frac{1}{2}\operatorname{osc}_w D_\delta\right] \leq \Gamma_s(p)\|a\|_r$, with $c = c_*$ (proved payment) or c_{12} (conjectured sharp). Proved for $p \leq 23$ (c_*), for $p \leq 59$ (c_{12}); (C_{L2}) is a phase-free sufficient condition. For comparison, the weaker common-point relaxation of the note has numerical critical constants $0.45, 0.43, 0.49, 0.50, 0.53, 0.57, 0.57, 0.61, 0.61$ ($p = 5, \ldots, 31$) against the needed 0.289 (`relax.py`, `ccrit*.out`; maximisers are $y^\star$ with $v = 0$).

6. Large p: relation mass is the Laplace transform of coverage

Lemma 8 (exact relation mass)**.** *For $S \subseteq \mathbb{Z}_p$ and $x \in \mathbb{R}^S_{\geq 0}$:* $\displaystyle\sum_{k\in\mathbb{Z}^S:\ \sum k_l l \equiv 0} \prod_l I_{|k_l|}(x_l) \;=\; Z(x) := \frac{1}{p}\sum_{\delta\in\mathbb{Z}_p} e^{\sum_l x_l \cos(2\pi l\delta/p)} = e^{\sum x_l}\ \frac{1}{p}\sum_\delta e^{-(Kx)_\delta}.$

(Use $e^{x\cos\vartheta} = \sum_k I_k(x) e^{ik\vartheta}$ and orthogonality.) Likewise, for any weighted product test with factor generating function $\hat\omega_l(\vartheta) = \sum_k \omega_l(k) e^{ik\vartheta}$, the absolute weight landing on the character m is $\frac{1}{p}\sum_\delta e^{-2\pi i m\delta/p} \prod_l \hat\omega_l(2\pi l\delta/p)$ (exact; with $|\omega|$ for worst-case phases).

Proposition 9 (exponential tilt: coverage ⇒ oscillation)**.** *Let $P(w) = \sum_{l\in S} b_l \cos(2\pi lw/p + \psi_l)$, $\lambda > 0$, $x = \lambda b$. Then, uniformly in the phases,* $\max P,\ -\min P \;\geq\; \frac{1}{\lambda}\log\big(2\prod_l I_0(x_l) - Z(x)\big)$. *In particular, with $E(x) = \sum_l (x_l - \log I_0(x_l))$ and $C(x) = \min_{\delta\neq 0}(Kx)_\delta$: if $E(x) \leq \ln(p/2)$ and $C(x) \geq E(x) + \ln 2$, then* $\max P, -\min P \geq \frac{1}{\lambda}\sum_l \log I_0(\lambda b_l)$.

Proof. $e^{\lambda \max P} \geq \mathbb{E}_w e^{\lambda P} = \sum_{k\in\mathrm{Rel}} \prod_l I_{k_l}(x_l) e^{ik_l\psi_l} \geq \prod I_0(x_l) - (Z(x) - \prod I_0(x_l))$ by Lemma 8; same for $-P$. For the second part, $Z \leq e^{\sum x}(\frac{1}{p} + e^{-C}) \leq e^{\sum x} e^{-E}$ and $\prod I_0 = e^{\sum x - E}$. □

In the relations case apply it to $P = D_\delta$ ($b_l = 2v_l s_l$): the IF bound holds at δ if $\frac{1}{\lambda}\log I_0(\lambda b_l) \geq c\,b_l s_l$ for each l. For a uniform cheap set of size $K = \kappa \ln p$ the certified fraction $2\log I_0(x)/x$ (with $\kappa(x - \log I_0 x) = 1$) and the coverage it requires, $\min_{\delta\neq 0} \frac{1}{K}\sum_{l\in S} \kappa_l(\delta) \geq 1 - \log I_0(x)/x$, are:

$\kappa = K/\ln p$	1.0	1.1	1.2	1.3	1.4	1.5	1.6
certified osc $/\sum b$	0.664	0.587	0.526	0.476	0.435	0.400	0.370
coverage needed	0.668	0.707	0.737	0.762	0.783	0.800	0.815

Needed: $2c\,\bar s \leq 0.577$ (c_*) or ≤ 0.367 (c_{12}). So at the programme scale $K^* \approx 1.3$–$1.5 \ln p$ the tilt certifies enough *under (P_{12})* ($K \leq 1.61\ln(p/2)$, coverage ≥ 0.82) and not under Theorem A ($K \leq 1.11\ln(p/2)$). The computed supports have uniform coverage 0.59–0.76 ($p \leq 101$), below 0.8, so the min-coverage form of Proposition 9 is not yet enough; the exact average $\frac{1}{p}\sum_\delta e^{-(Kx)_\delta}$ of Lemma 8 is what must be estimated. By contrast a linear Riesz product cannot be controlled by min-coverage at all: the same computation needs coverage $\geq (1+\varepsilon)\log(1+\varepsilon)/\varepsilon > 1$.

7. Relation weights of weighted product tests (requested by the main thread)

In the relations case $\gamma_l = 0$ and the quadratic factor is exact: $w_l = (\cos u_l - 2^{-1/2})^2$, so $1-\varepsilon+\varepsilon w_l = 1 - \sqrt{2}\,\varepsilon\cos u_l + \frac{\varepsilon}{2}\cos 2u_l$. Its monomials sit at $\pm l$ (weight $\varepsilon/\sqrt{2}$) and $\pm 2l$ (weight $\varepsilon/4$): off-zero mass $(\sqrt{2}+\frac{1}{2})\varepsilon = 1.91\varepsilon$ per factor, against ε for the linear factor $1-\varepsilon\cos u_l$. Detection per unit b_l: $\varepsilon/\sqrt{2}$ (quadratic), $\varepsilon/2$ (linear); payment per unit b_l: $c_* s_l \le 0.289$, i.e. at most 40.8% of the quadratic and 57.7% of the linear detection. The relations that leak are $\sum_l k_l l \equiv 0$ (normalisation W_0) and $\sum_l k_l l \equiv \pm m$, $m \in X$ (leak Λ_m), with $k_l \in \{0, \pm 1\}$ (linear) or $\{0, \pm 1, \pm 2\}$ (quadratic); each relation leaks $\prod_l \omega(|k_l|)$. Per-feature surplus $\big[\det -\Lambda_m - (1+W_0)c_* s_m(\delta)\big]/\det$, minimised over m, maximised over ε and binding δ (`relq2.out`):

p	5	7	11	13	17	19	23	29	31	37	41	47
linear	−5	−68	+56	+26	+46	+47	+6	+27	+33	−3	+5	−38
quadratic	−79	−168	+60	−45	−11	+1	−63	−50	−26	−72	−47	−125
p	53	59	61	67	71	73	79	83	89	97		
linear	+27	+20	+22	−3	−1	−3	+6	−3	+12	+8		
quadratic	−36	−38	−38	−67	−74	−72	−62	−54	−52	−51		

(in % of detection; worst-case phases). So in the relations case product tests are the wrong tool: the quadratic one leaks more than the linear one (more monomials, coefficient-2 relations), both fail at many primes, and Parseval, which sees no relation at all, closes every one of these supports (Corollary 4(c)). For the multi-T setting of the note's quadratic theorem the same exact formula applies on $\mathbb{Z}_p^2$: leak at $\chi = p^{-2}\sum_{\zeta\in\mathbb{Z}_p^2} e^{-2\pi i\langle\chi,\zeta\rangle/p}\prod_l \hat{\omega}_l(\zeta)$, where $\hat{\omega}_l(\zeta)$ is the generating function of w_l's monomials $2l(\pm\mathbf{1}_T \pm \mathbf{1}_{T'})$ evaluated at ζ. A relation in $\mathbb{Z}_p^2$ needs one in each coordinate, so the single-T (diagonal) assignment is the relation-richest; within the 39% budget, the leakage to fit is $\sqrt{2}\max_m \Lambda_m/\varepsilon + W_0$ (quadratic) in units of detection.

8. Literature

No off-the-shelf result applies. Chowla's cosine problem (Bedert, arXiv:2509.05260, polynomial lower bounds for $-\min$ of 0/1 cosine sums) and its one-sided variants concern many frequencies and bounds sublinear in $\sum b$; we need bounds linear in $\sum b$ for $|S| \asymp \log p$ with arbitrary phases, which fail in general (flat polynomials). Sidon-set theory (Pisier; Bourgain, Ann. Inst. Fourier 35 (1985); Hare–Yang, arXiv:1808.03128) is qualitative and gives no constants for $|S| \asymp \log p$ in $\mathbb{Z}_p$. The arithmetic input that replaces dissociation here is Lemma 8.

9. Precise remaining gap

(G1) Prove (P_{12}) for complex atoms (a single-feature inequality). This extends the closure from $p \le 23$ to $p \le 59$ unconditionally.

(G2) For $p > 23$ (resp. 59), the only obstruction is cheap interference spread over most of F; prove (C_{L2}) (numerically true for $29 \le p \le 61$ with $\ge 15\%$ margin) or an averaged-coverage version of Proposition 9.

(G3) For $\ln p \gtrsim 11$, programme-scale cheap sets exceed Parseval's threshold; Proposition 9 with the averaged form of Lemma 8 and (P_{12}) is the candidate argument (at $K = 1.5\ln p$ it certifies 0.400 against 0.367 needed).

Files: `scratchpad/fork01/`: `relax.py, ccrit.py` (common-point relaxation), `sigma.py, sigsupp.py` (plain one-sided constants), `cl2f.py, cl2g.py` (C_{L2}), `c12.py` (sharp single-T payment), `softmax.py` (Lemma 8 check and tilt bound), `relq.py, relq2.py` (product-test relation weights), `supports_*.json`.